Online Book Store Database System

A Group Experiment in Relational Systems

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Abstract

This project implements a relational database system for an online bookstore. The database is designed to manage books, authors, publishers, customers, and transactions such as orders and payments. It also supports categorization of books and customer reviews. The goal of this project is to demonstrate proper database design, normalization, and implementation using SQL. The system models real-world bookstore operations and allows for efficient querying and data management. The database is implemented using SQLite and populated using Python scripts with both real-world inspired data and synthetic test data.

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Introduction

This project focuses on designing and implementing a database for an online bookstore system. The database stores information about books, authors, publishers, customers, and transactions such as orders and payments. The motivation for this database is to simulate a real-world e-commerce bookstore system. Such a system is

useful for managing inventory, tracking customer purchases, organizing books by categories, and enabling customer feedback through reviews.

The intended users of this database include:

- Store administrators managing inventory and orders
- Customers purchasing books and leaving reviews
- Analysts querying sales and customer behavior

This system demonstrates how relational databases can be used to model complex real-world systems with multiple relationships.

Database Design

Requirements, Assumptions, and Constraints

User Requirements

The system must:

- Store customer information
- Store books and publisher details
- Support multiple authors per book
- Categorize books
- Record orders and payments
- Allow reviews for books
- Maintain relationships between all entities

Business Rules

- Each customer has a unique ID and email
- Each book has a unique ISBN
- Each book belongs to one publisher
- Books can have multiple authors and categories
- Orders belong to one customer
- Orders can contain multiple books

- Each order has one payment
- Customers can leave multiple reviews

Assumptions

- Publisher country defaults to USA during data import
- ISBN values are generated programmatically
- Customer data is synthetic for testing
- Reviews only store ratings (no text)
- Location data is simplified (city/state only)

Constraints

- Primary keys on all tables
- Foreign key constraints enforce relationships
- Unique constraints:
    - email (customers)
    - ISBN (books)
    - category name
- Composite keys for bridge tables
- NOT NULL constraints for essential fields

Entity-Relationship Diagram

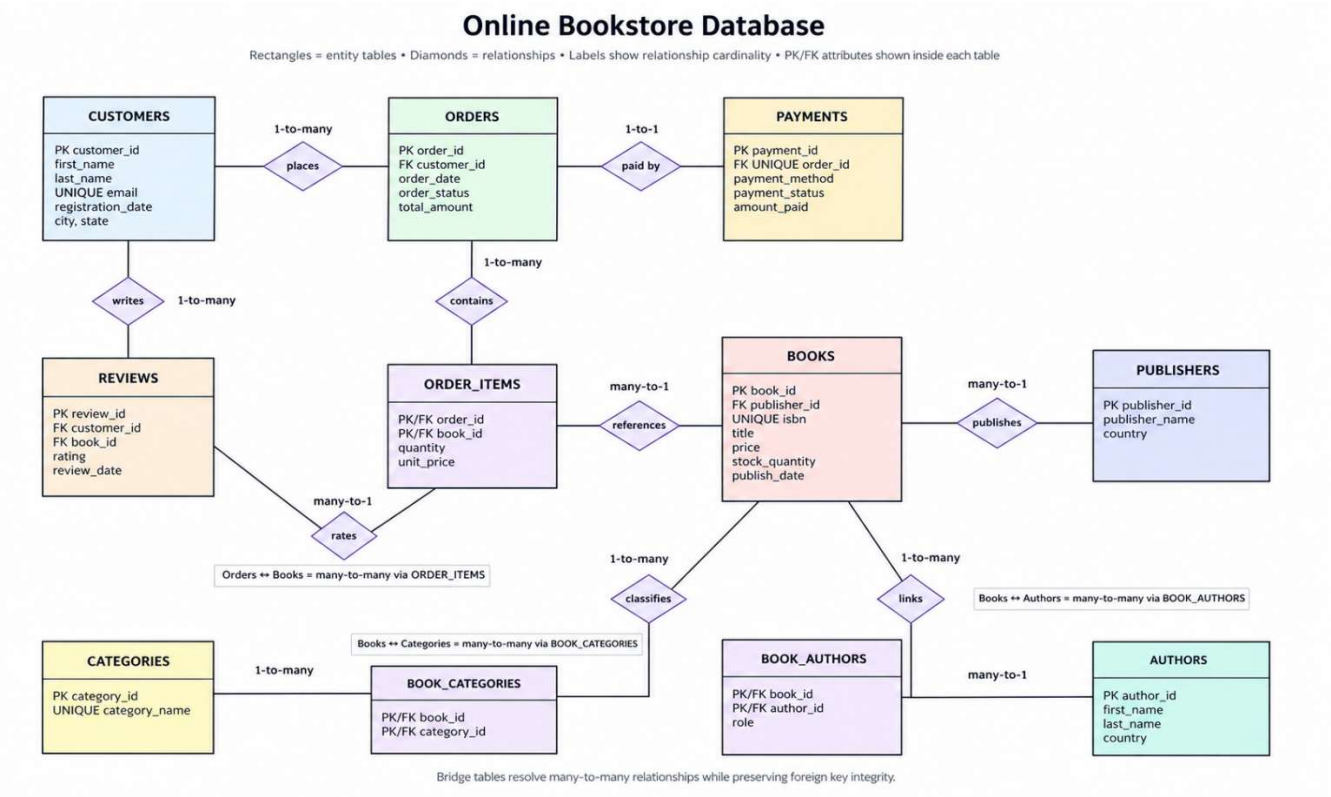

Table Schemas and Functional Dependencies(FDs)

Customers

- PK: customer_id
- Attributes: first_name, last_name, email, registration_date, city, state

FDs:

- customer_id → all attributes
- email → all attributes

Publishers

- PK: publisher_id

FD:

- publisher_id → publisher_name, country

Books

- PK: book_id

- FK: publisher_id

FDs:

- book_id → all attributes
- isbn → all attributes

Orders

- PK: order_id
- FK: customer_id

FD:

- order_id → all attributes

Authors

- PK: author_id

FD:

- author_id → all attributes

Categories

- PK: category_id

FDs:

- category_id → category_name
- category_name → category_id

Payments

- PK: payment_id
- FK: order_id (UNIQUE)

FDs:

- payment_id → all attributes
- order_id → all attributes

Order_items

- PK: (order_id, book_id)

FD:

- (order_id, book_id) → quantity, unit_price

Reviews

- PK: review_id

FD:

- review_id → all attributes

Book_authors

- PK: (book_id, author_id)

FD:

- (book_id, author_id) → role

Book_categories

- PK: (book_id, category_id)

FD:

- Only trivial dependencies

Form

All tables are in **Third Normal Form(3NF).**

Database Implementation

Platform and Tools

- SQLite (database engine)
- Python (data loading scripts)
- Pandas / NumPy (data handling)
- HTML dashboard (visualization layer)

Data Definition

Tables are created using SQL **CREATE TABLE** statements implemented in Python.

Example:

```
CREATE TABLE customers (
  customer_id INTEGER PRIMARY KEY,
```

```
  first_name TEXT NOT NULL,
  last_name TEXT NOT NULL,
  email TEXT UNIQUE NOT NULL,
  registration_date TEXT NOT NULL,
  city TEXT NOT NULL,
  state TEXT NOT NULL
);
```

Sample Insert Statements

Customers

```
INSERT INTO customers VALUES
(1, 'John', 'Smith', 'john.smith@gmail.com', '2025-01-15', 'Dallas', 'TX');

INSERT INTO customers VALUES
(2, 'Jane', 'Doe', 'jane.doe@gmail.com', '2025-02-03', 'Chicago', 'IL');
```

Publishers

```
INSERT INTO publishers VALUES
(1, 'Penguin Random House', 'USA');

INSERT INTO publishers VALUES
(2, 'HarperCollins', 'USA');
```

Books

```
INSERT INTO books VALUES
(1, 1, '978-123456789', 'Database Systems Concepts', 59.99, 25);

INSERT INTO books VALUES
(2, 2, '978-987654321', 'Introduction to SQL', 39.99, 40);
```

Authors

```
INSERT INTO authors VALUES
(1, 'Henry', 'Korth', 'USA');

INSERT INTO authors VALUES
(2, 'Abraham', 'Silberschatz', 'USA');
```

Categories

```
INSERT INTO categories VALUES
(1, 'Computer Science');

INSERT INTO categories VALUES
(2, 'Database');
```

Orders

```
INSERT INTO orders VALUES
(1, 1, '2025-04-01', 'Completed', 59.99);

INSERT INTO orders VALUES
(2, 2, '2025-04-02', 'Pending', 39.99);
```

Payments

```
INSERT INTO payments VALUES
(1, 1, 'Credit Card', 'Paid', 59.99);

INSERT INTO payments VALUES
(2, 2, 'PayPal', 'Pending', 39.99);
```

Order_items

```
INSERT INTO order_items VALUES
(1, 1, 1, 59.99);

INSERT INTO order_items VALUES
(2, 2, 1, 39.99);
```

Reviews

```
INSERT INTO reviews VALUES
(1, 1, 1, 5, '2025-04-05');

INSERT INTO reviews VALUES
(2, 2, 2, 4, '2025-04-06');
```

Book_authors

```
INSERT INTO book_authors VALUES
(1, 1, 'Author');
```

INSERT INTO book_authors VALUES
(1, 2, 'Author');

Book_categories

INSERT INTO book_categories VALUES
(1, 1);

INSERT INTO book_categories VALUES
(1, 2);

Table Output Post Insert

SELECT * FROM customers;

customer_id	first_name	last_name	registration_date	city	state
1	Grace	Johnson	2025-11-08	New York	NY
2	Charlie	Taylor	2025-06-02	Philadelphia	PA
3	Alice	Anderson	2025-09-14	Chicago	IL
4	Charlie	Brown	2025-07-18	San Jose	CA
5	Alice	Taylor	2026-01-13	San Diego	CA
6	Diana	Doe	2025-08-05	San Diego	CA
7	Charlie	Johnson	2025-07-12	Los Angeles	CA
8	Charlie	Davis	2026-03-26	Philadelphia	PA
9	Hank	Taylor	2025-08-24	Chicago	IL
10	Eve	Moore	2025-04-22	San Antonio	TX

SELECT * FROM publishers;

publisher_id	publisher_name	country
1	Doubleday	USA
2	Putnam Pub Group	USA
3	Workman Pub Co	USA
4	Natl Pr Books	USA
5	Amy Spangler	USA
6	Excalibur Press	USA
7	Health Communications Inc	USA
8	VSP Books	USA

publisher_id	publisher_name	country
9	Random House	USA
10	Oxmoor House	USA

SELECT * FROM books;

book_id	publisher_id	price	stock_quantity
1	1	8.79	44
2	2	4.99	34
3	3	4.99	26
4	4	4.99	96
5	5	5.32	15
6	6	6.06	20
7	7	4.99	4
8	8	19.96	90
9	9	4.99	61
10	10	12.98	97

SELECT * FROM authors;

author_id	first_name	last_name	country
1	Larry	Colton	USA
2	Doris	Grumbach	USA
3	Joseph C.	Piscatella	USA
4	Paul D.	Davis	USA
5	Amy	Spangler	USA
6	Bo	Short	USA
7	Jack (COM)Hansen	Canfield	USA
8	Mattie J. T.	Stepanek	USA
9	Vassily	Aksyonov	USA
10	Michael Henry	Heim	USA

SELECT * FROM categories;

category_id	category_name
1	History

category_id	category_name
2	General
3	Fiction
4	Cooking
5	Reference
6	(empty)
7	Self-help
8	Personal Growth
9	Self-Esteem
10	Poetry

SELECT * FROM orders;

order_id	customer_id	order_date	order_status	total_amount
1	62	2026-03-27	Delivered	14.97
2	16	2025-10-30	Pending	19.96
3	69	2025-11-10	Shipped	19.96
4	147	2025-12-31	Delivered	19.25
5	56	2026-02-15	Shipped	4.99
6	108	2025-12-13	Cancelled	21.16
7	79	2026-04-12	Pending	21.83
8	38	2025-08-13	Pending	15.87
9	10	2025-05-21	Pending	16.08
10	29	2026-02-15	Shipped	43.96

SELECT * FROM payments;

payment_id	order_id	payment_method	payment_status	amount_paid
1	1	Credit Card	Completed	14.97
2	2	PayPal	Completed	19.96
3	3	Credit Card	Completed	19.96
4	4	PayPal	Pending	19.25
5	5	PayPal	Completed	4.99
6	6	PayPal	Completed	21.16
7	7	Credit Card	Pending	21.83
8	8	Bank Transfer	Pending	15.87

payment_id	order_id	payment_method	payment_status	amount_paid
9	9	Bank Transfer	Completed	16.08
10	10	Bank Transfer	Completed	43.96

SELECT * FROM order_items;

order_id	book_id	quantity	unit_price
1	180	3	4.99
2	75	4	4.99
3	13	4	4.99
4	94	1	19.25
5	408	1	4.99
6	414	4	5.29
7	195	1	21.83
8	414	3	5.29
9	301	3	5.36
10	63	4	10.99

SELECT * FROM reviews;

review_id	customer_id	book_id	rating	review_date
1	105	435	1.4	2025-07-18
2	31	238	7.1	2025-09-12
3	10	474	1.7	2025-08-31
4	95	220	1.2	2025-06-20
5	115	459	2.8	2025-07-29
6	70	294	9.6	2025-05-05
7	23	71	7.2	2026-01-08
8	45	469	0.3	2026-03-02
9	16	219	9.8	2025-10-21
10	109	78	7.5	2025-09-03

SELECT * FROM book_authors;

book_id	author_id	role
1	1	Author
2	2	Author
3	3	Author
4	4	Author
5	5	Author
6	6	Author
7	7	Author
8	8	Author
9	9	Author
9	10	Author

SELECT * FROM book_categories;

book_id	category_id
1	1
1	2
2	3
2	2
3	4
3	5
4	6
5	6
6	6
7	7

Data Manipulation

Example Queries:

1. Get all books with publisher names

```
SELECT b.title, p.publisher_name
FROM books b
JOIN publishers p ON b.publisher_id = p.publisher_id;
```

2. Count books per category

```
SELECT c.category_name, COUNT(*)
FROM categories c
JOIN book_categories bc ON c.category_id = bc.category_id
GROUP BY c.category_name;
```

3. Top expensive books

```
SELECT title, price
FROM books
ORDER BY price DESC
LIMIT 10;
```

4. Customers per state

```
SELECT state, COUNT(*)
FROM customers
GROUP BY state;
```

5. Books with multiple authors

```
SELECT book_id, COUNT(author_id)
FROM book_authors
GROUP BY book_id
HAVING COUNT(author_id) > 1;
```

Example Update Queries:

```
UPDATE books
SET stock_quantity = stock_quantity - 1
WHERE book_id = 1;

UPDATE customers
SET city = 'Baton Rouge'
WHERE customer_id = 5;
```

Conclusion

This project demonstrates the design and implementation of a normalized relational database for an online bookstore. The system successfully models real-world relationships between books, authors, publishers, and customers.

Future improvements could include:

- Adding a web-based frontend.
- Expanding review functionality (text, sentiment analysis).
- Implementing real payment processing logic.
- Adding recommendation systems.

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